

First year Aerospace Engineering

Mathematics - Calculus II

Marcos Carrera Valle

Indefinite Integrals

Find the following integrals:

A) $\int \sec(x)(\sec(x)+\tan(x)) \, dx$

B) $\int \cos(x) / (1-\cos^2(x)) \, dx$

C) $\int ce^{ax+b+d} \, dx$

D) $\int (x-5) / (2x^2+x-1) \, dx$

Complicated Integrals

Solve the following integrals:

A) $\int \sin(2x/3) \cdot \sin^2(x/3) dx$

B) $\int (4\cos^2(2x))/(\sin^2(4x)) dx$

Definite Integrals

Solve the following definite integrals:

A) $\int_{\pi/4}^{\pi/3} 1 + \tan^2(x) \, dx$

B) $\int_{\pi/12}^{\pi} 8 \sin(x) \cos(x) \, dx$

C) $\int_{\pi/12}^{\pi} 3(x+2)^2 - (x^2 - 9x - 36) / (x+3) \, dx$

Partial Fractions

Solve the following integrals:

A) $\int x / (x^3 - 5x^2 + 8x - 4) dx$

B) $\int (2x^4 + x + 8) / (x^3 + 9x) dx$

C) $\int (3x^2 - 10) / (x^2 - 4x + 4) dx$

Harder Integrals

Solve the following integrals:

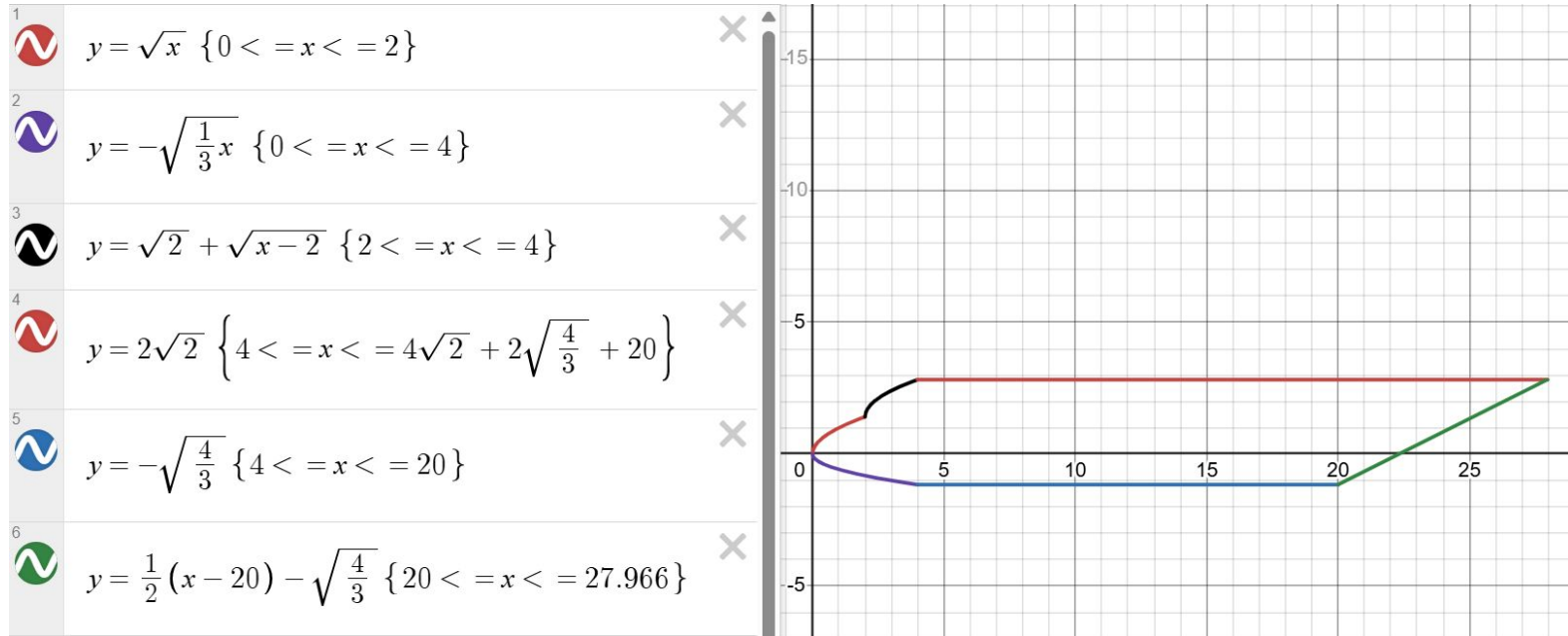
A) $\int x^2 e^x dx$

B) $\int [\ln(x)]^2 dx$

C) $\int [\ln(x)]^2 / x \, dx$

Volumes of revolution (HARD)

Find the volume of the aircraft defined by the following formulae (use volumes of revolution around the x axis)



Indefinite Integrals

Find the following integrals:

A) $\int \sec(x)(\sec(x)+\tan(x)) dx$
 $\tan(x)+\sec(x)+C$

B) $\int \cos(x) / (1-\cos^2(x)) dx$
 $-\csc(x) + C$

C) $\int ce^{ax+b+d} dx$
 $c*a*e^{ax+b+d}*x + C$

D) $\int (x-5) / (2x^2+x-1) dx$
 $2\ln(x+1) - (3/2)*\ln(2x-1)$

Complicated Integrals

Solve the following integrals:

A) $\int \sin(2x/3) \sin^2(x/3) dx$

$$-\frac{3}{4} \cos(2x/3) + \frac{3}{16} \cos(4x/3) + C$$

B) $\int (4\cos^2(2x))/(\sin^2(4x)) dx$

$$-\frac{1}{2} [\csc(4x) + \cot(4x)] + C$$

Definite Integrals

Solve the following definite integrals:

A) $\int_{\pi/4}^{\pi/3} 1 + \tan^2(x) \, dx$
 $-1 + \sqrt{3}$

B) $\int_{\pi/12}^{\pi} 8 \sin(x) \cos(x) \, dx$
 $-2 + \sqrt{3}$

C) $\int_{\pi/12}^{\pi} 3(x+2)^2 - (x^2 - 9x - 36) / (x+3) \, dx$
 $70\frac{1}{2}$

Partial Fractions

Solve the following integrals:

A) $\int x / (x^3-5x^2+8x-4) dx$
 $\ln((x-1)(x-2)) - 2 / (x-2) + C$

B) $\int (2x^4+x+8) / (x^3+9x) dx$
 $x^3+8/9 \ln(x)-85/9\ln(x^2+9)+1/3\tan^{-1}(x/3)+C$

C) $\int (3x^2-10) / (x^2-4x+4) dx$
 $3/2 x^2 + 12\ln(x-2)-2/(x-2)^1+C$

Harder Integrals

Solve the following integrals:

A) $\int x^2 e^x dx$
 $e^x [x^2 - 2x + 2] + C$

B) $\int [\ln(x)]^2 dx$
 $x(\ln(x))^2 - 2x\ln(x) + 2x + C$

C) $\int [\ln(x)]^2 / x dx$
 $[\ln(x)]^3 / 3 + C$

Volumes of revolution (HARD)

Find the volume of the aircraft defined by the following formulae

$$V = 274.13 \text{ m}^3$$

